

Perpetual Futures: Mechanics, History and Purpose

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The Commodity Futures Trading Commission (CFTC) recently cleared the way for a new kind of contract — the perpetual future — to trade in the United States, opening the door for platforms like Kalshi and Coinbase to offer them to American investors. The decision didn't sit well everywhere: the CME Group, the country's largest futures exchange, promptly filed suit to block it.² Perpetual futures have quietly dominated crypto trading outside the U.S. for the better part of a decade, and they're the primary way traders get leveraged exposure to Bitcoin. Kalshi claims offshore perpetuals grew from \$28 trillion in annual volume in 2023 to more than \$90 trillion in 2025.³ With the recent CFTC regulatory blessing in hand, U.S. platforms plan to push the same structure well beyond crypto, into individual stocks, stock indexes, commodities, and more.

Perpetual futures have been very much in the news recently (excellent articles have appeared in the [Buttonwood](#) column of the Economist, the [FT](#), the [WSJ](#) and Matt Levine's Bloomberg *Money Stuff* column). In conversations with friends and clients, we've noticed real interest in understanding how these contracts actually work, along with some curiosity about their pros and cons relative to traditional futures. So, let's start with a quick refresher on how an ordinary futures contract works.

What is a Futures Contract?

A futures contract is a standardized agreement to buy or sell an underlying asset — a stock index, a U.S. government bond, a barrel of oil — at a predetermined price on a specified future date, traded on an exchange that guarantees the trade through a clearinghouse. Both parties are obligated to fulfill the contract — the buyer to purchase, the seller to deliver — unless the position is closed out before expiration, and settlement can be either physical delivery of the asset or cash settlement based on its price at expiration. Each day, gains and losses are settled through variation margin: cash flows between the two parties' margin accounts that mark the position to the current market price, rather than waiting until expiration to realize the full profit or loss.

Futures markets have existed in organized form since the 1800s, making them one of the oldest financial derivatives still in wide use today. A key part of their appeal is leverage — since only a fraction of a contract's notional value (margin) is required upfront, traders can gain outsized exposure to price moves relative to the capital they've committed, which cuts both ways: amplifying gains, but also losses.

Ordinarily, contracts are listed with expirations spread out every one to three months, with most of the liquidity concentrated in the contract with the nearest expiration. A trader who wants to maintain a position beyond the expiration of their current contract has to "roll" it — simultaneously closing the expiring contract and opening a new one further out. Cash-settled, traditionally structured Bitcoin

¹ This is not an offer or solicitation of investment services or financial advice. **Past returns are not indicative of future performance.** Thanks to Jerry Bell, Aaron Brown, Aneet Chachra, Richard Dewey, Nicholas Dunn, Nikita Fadeev, David Heatley, Agustin Lebron, Steve Mobbs, Ricky Moezinia, Craig Newbold, Vladimir Ragulin, Vijay Sharma and Lewis Tuff for their helpful comments.

² One of the CME's arguments is that, because both sides exchange periodic funding payments, perpetuals meet the Dodd-Frank definition of a swap, not a future.

³ Kalshi (2026).

futures were introduced on the CME at the end of 2017, and continue to trade there today with decent liquidity (recently about \$10 billion per day).

What's different about Perpetual Futures?

Perpetual futures, in their current form, got their start in mid-2016 (a year and a half earlier than the CME BTC futures) as a derivative on Bitcoin – and were invented, at least in part, to get around this exact rolling problem.

Perpetual futures are futures contracts with no expiration date at all, which lets traders hold leveraged long or short positions indefinitely without ever needing to roll into a new contract. But without an expiration date, there's no "date with destiny" forcing the contract's price to converge with the spot price the way a normal futures contract eventually must. So perpetual futures platforms use a "funding rate" instead: a periodic payment exchanged directly between long and short position holders, designed to keep the perpetual's price anchored to the spot price of the underlying asset. It's called a funding rate because it represents the ongoing cost for the long side keeping the position open.

The logic is simple. Suppose Bitcoin's spot index – a composite of prices across major spot exchanges – stands at \$60,000, while buying pressure has the perpetual trading at \$60,060, a premium of 0.10%. That premium produces a funding rate of about 0.05% (the formula shaves off a small buffer). For as long as the premium lasts, every long pays 0.05% of their position's value and every short receives it – \$30 per Bitcoin of exposure, three times a day, for as long as the premium lasts. That steady drip is what connects the price of the futures to the price of spot Bitcoin; an arbitrageur can buy Bitcoin spot at \$60,000, short the perpetual at \$60,060, and – hedged against the market – collect the funding until the gap closes. Their arbitrage-drive selling of the perpetual – or longs simply declining to keep paying the funding rate – is what pushes the perpetual back toward spot.

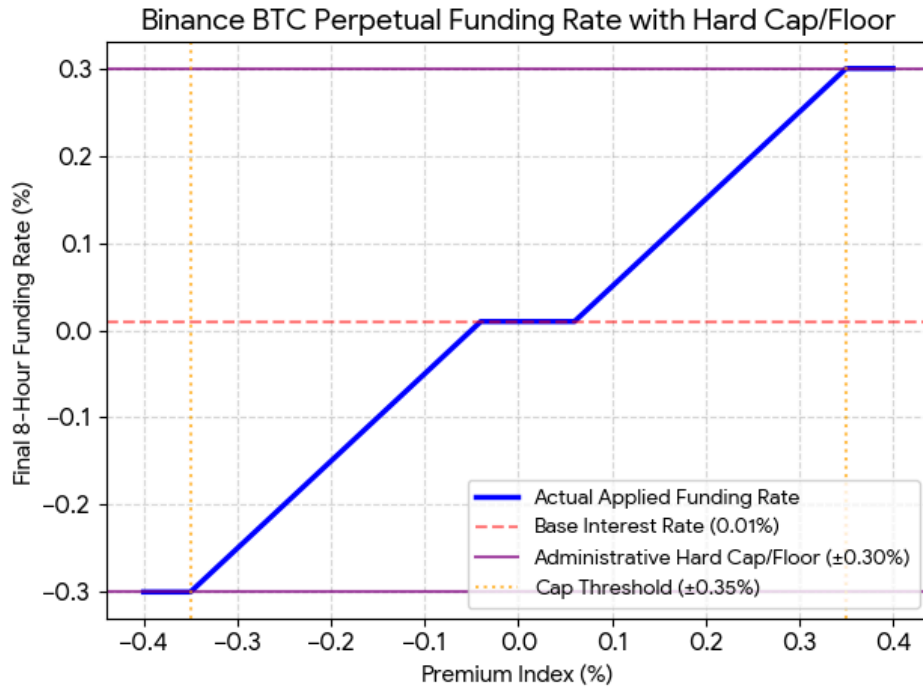
On the Binance platform – the largest venue for crypto perpetuals by a wide margin – the funding rate on the BTC perpetual is calculated as:

$$\text{Funding Rate} = \text{Premium Index (P)} + \max(-0.05\%, \min(0.01\% - P, 0.05\%))$$

where P is the percentage gap between the perpetual's price and Bitcoin's spot price, averaged by Binance over the trailing eight hours. When the funding rate is positive, longs pay shorts; when negative, shorts pay longs – with Binance acting as the middleman that nets the payments out. The 0.01% term is a "small" fixed interest-rate component (Binance's default of 0.03% per day, split across the three eight-hour settlements), and the "max/min" clamp keeps a modest premium from moving the rate at all: as long as P stays within about five basis points of 0.01%, the funding rate holds at the flat 0.01% baseline regardless of what the premium is actually doing. Only once the premium moves far enough away does it start to pass through into the realized rate, net of that five-basis-point buffer.

There's also a hard ceiling on how far the rate can move in either direction over a single eight-hour window. For Binance's standard USDT-margined BTC contract, that cap currently sits at 0.30% per eight hours.⁴ The chart below shows the relationship between the futures vs spot price premium (discount) and the Funding rate.

⁴ Coin-margined – "inverse" – contracts, where traders post actual Bitcoin as collateral, get a wider cap, often stretched out to around 0.75%, to account for the added volatility of the collateral itself.



To see how the formula with the full set of constraints plays out: suppose the premium spikes from 0% to 0.5% during a short squeeze. Plugging into the formula, funding rate = $\min(0.3\%, 0.5\% + \max(-0.05\%, \min(0.05\%, 0.01\% - 0.5\%))) = 0.3\%$. Note that 0.3% per eight hours annualizes to 2600%.⁵

⁵ During periods of extreme stress, exchanges sometimes shorten the funding interval from eight hours down to one hour to let the mechanism react faster. When that happens, the interest-rate component doesn't get divided by eight – it gets rebuilt on an hourly basis (Binance's daily 0.03% baseline divided by 24 rather than three, so I drops from 0.01% to about 0.00125%), and the hard cap is scaled down as well, though not simply divided by eight either.

A Brief History of Non-Expiring Futures

The conventional origin story credits BitMEX, founded by Arthur Hayes,⁶ with the current format – a 2016 innovation combining no-expiration contracts with the funding-rate mechanism to keep price anchored, purpose-built for Bitcoin. But long before Hayes, the Chinese Gold and Silver Exchange Society (CGSE) of Hong Kong had been running a physical, open-outcry version of the same idea for nearly a century. Founded in 1910, the CGSE’s flagship 99 Tael Gold Contract never expired: every business day at a designated Declaration Time, longs could demand physical delivery of gold bars the next day, and shorts could declare their intent to deliver. Because delivery could be demanded at any time, the contract stayed tethered to the spot price of gold directly – it wasn’t really a funding rate doing the work of keeping price in line, but the ever-present threat of delivery. A daily rate for financing overnight positions was set through an auction among members. The mechanism was remarkably successful: for much of the 20th century, it helped make Hong Kong one of the top three or four gold-trading hubs in the world. The CGSE was restructured in 2025 into the Hong Kong Gold Exchange (HKGX), and the 99 Tael contract, undated mechanism intact, is still traded today.

In the academic literature, these are sometimes called “undated futures” or “deferred spot contracts” which do have some characteristics in common with the modern crypto perpetual. Adam Gehr’s 1988 paper on the topic argued that undated futures reduce basis risk and concentrate liquidity into a single contract rather than splintering it across several expirations – essentially the same case made for crypto perpetuals today (Gehr, 1988).

Yale economist Robert Shiller is also credited with contributing to the idea, and the “perpetual futures” moniker, via his 1993 paper on measuring asset values for cash settlement in illiquid derivative markets (Shiller, 1993). Shiller’s “perpetual futures” concept was aimed at giving investors exposure to assets – like real estate or macroeconomic indices – that don’t have liquid, continuously-traded spot markets. Notably, the premium/discount versus spot component of the daily funding-rate described above wasn’t part of his proposal.

Funding rate history

Over the past six years, the funding rate on Binance – the largest exchange trading crypto perpetuals – has averaged about 14% per annum, against a U.S. T-bill rate that has averaged around 3%, a gap of 11%. This is about three times larger than the financing spread on CME-listed BTC futures⁷ and 7 - 20 times larger than the analogous financing spread on traditional futures on bond and equity indexes – which has averaged between 0.5% to 1.5%. That’s a striking difference, and one that says a lot about how much more expensive it has been, historically, to get leveraged long exposure to Bitcoin on offshore platforms than on traditional, onshore regulated exchanges.

⁶ “Following a massive rise to prominence in the digital asset space, Hayes stepped down as CEO of BitMEX in 2020 amid regulatory pressures from the U.S. government. In 2022, he pleaded guilty to violating the Bank Secrecy Act... He later received a pardon from President Donald Trump.” (source: Wikipedia)

⁷ Based on the return of the Bloomberg index (ticker BTCFTR) of the total collateralized return of CME BTC futures versus the return of BTC.

Binance BTC Perpetual Funding Rates by Year (source: Binance’s public historical data repository)

Year	Avg. 8-Hr Rate	Annual-ized Fund-ing Rate	Market Regime & Context
2020	+0.016%	18.8%	Post-halving awakening: funding sat near the 0.01% floor for most of the year, drifting into mild premiums during the Q4 breakout past \$20,000.
2021	+0.028%	35.9%	Peak bull frenzy: retail leverage surged, and longs paid over 40% annualized just to sit in positions. Arbitrageurs made fortunes buying spot and shorting perps.
2022	+0.004%	4.3%	Crypto winter: deleveraging events (Terra/Luna, Celsius, FTX) pushed the average below the 0.01% baseline for the first time, on structural panic and one-sided short pressure.
2023	+0.007%	8.2%	Institutional accumulation: the market recovered and funding snapped back to its 0.01% gravity point, creeping up gently on Q4 ETF anticipation.
2024	+0.011%	12.7%	ETF-inflow bull run: Bitcoin set new highs. Q1 was volatile, with large positive spikes (+0.10% to +0.25%) before moderating into a steadier baseline.
2025	+0.005%	5.3%	Mature-market consolidation: deep institutional liquidity compressed structural premiums, anchoring rates firmly near the 0.01% baseline.
Avg		13.7%	Average U.S. T-bill rate over the period was 3.1%.

Academic work backs up the broad pattern here. He *et al.* (2023) find that crypto perpetual prices, driven by intense retail momentum, routinely detach from spot – with annualized deviations of 60% to 100% – far exceeding anything seen in traditional financial markets.⁸ The reason the arbitrage doesn’t compress this away faster is that executing the offsetting basis trade means facing real margin-liquidation risk, managing which can require both risk capital and margin capital. The complicated funding rate formulae also tend to exacerbate the difficulties of effectively arbitraging spot vs perpetual futures. So the mispricing can persist well outside anything close to theoretical bounds.

Other Features: Leverage, Liquidation, and Insurance Funds

Leverage of 20x to 100x is entirely normal on crypto perpetuals – meaning a trader might post as little as 1% to 5% of a position’s notional value as initial margin. Exchanges set a maintenance margin threshold (commonly around 0.5%, sometimes higher) and, unlike traditional brokers, don’t call for more collateral if it’s breached – they simply liquidate the position automatically. The mechanism, known as auto-liquidation, can produce brutal losses in a fast market – one news article notes that a surprise announcement of 100% tariffs on China this past October triggered a selloff that wiped out more than \$19 billion in leveraged crypto positions in a matter of hours.

When a position is auto-liquidated, the exchange typically keeps a liquidation fee out of whatever margin remains, funneling it into an insurance fund meant to cover shortfalls in cases where a liquidated position’s losses exceed its remaining margin.⁹ It’s worth being clear-eyed about what that fund

⁸ A fixed-maturity future’s price is anchored by expiration-date arbitrage; a perpetual has no such date, so that logic doesn’t apply. He *et al.* gets around this with “random-maturity arbitrage.” Since traders exit at a time of their own choosing rather than a fixed date, a no-arbitrage price still exists under that weaker requirement – exact without trading costs, a band once costs are included. That band both benchmarks fair value and prescribes the trade that should correct any drift from it.

⁹ If the insurance fund runs dry, money is taken from counterparties with winning positions to fund the shortfall.

actually is, legally speaking: despite being marketed as a user-protection mechanism, it isn't anything like FDIC insurance. Binance's own terms state that once the fund exceeds a required minimum, "any funds in excess of the required minimum may be deployed by Binance for other purposes as it considers appropriate in its sole discretion." The Celsius and FTX bankruptcies are the relevant precedent here: courts ruled that because each platform's terms of use gave it ownership and right-of-use over customer assets and pooled funds, those assets belonged to the bankruptcy estate, not to individual users, once the platform failed. If a major centralized exchange collapsed tomorrow, its insurance fund would almost certainly be frozen and folded into the general creditor pool.¹⁰

Futures Industry Reaction

"Perps are Prediction Markets. Prediction Markets are Perps"

– X post by John Wang, head of Crypto at Kalshi August 2025)

Tarek Mansour, Kalshi's chief executive, called perps "the purest form of trading" (FT, July 19, 2026). The traditional futures world doesn't seem convinced nor is it thrilled that perpetuals have come to U.S. markets. Beyond CME's lawsuit and CEO Terry Duffy's public objections,¹¹ other major exchanges are watching cautiously rather than rushing in. Stuart Williams, chief operating officer at ICE, called perpetual futures "riskier than their traditional counterparts," and while ICE has no current plans to launch its own, he said the company is "going back out" to gauge institutional demand – noting that "not a single customer has asked us for perps." NASDAQ is reportedly waiting for more regulatory clarity before acting, while Rob Hocking, CBOE's global head of derivatives, has said CBOE is considering converting its existing Bitcoin and ether futures – which currently list expirations out as far as a decade – into true perpetuals, now that the CFTC has said it will allow registered U.S. platforms to relist undated-style contracts as perpetual futures outright.

Pros and Cons versus other forms of leverage: same risk, higher price?

Proponents of perpetuals point to the obvious convenience: no rolling, and by concentrating trading into a single contract rather than splitting it across several listed expirations, liquidity pools more deeply in one place.

Critics counter that one- or three-month rolling futures contracts work perfectly well for an asset like Bitcoin. Indeed, regulated Bitcoin futures on the CME have functioned smoothly, co-existing with offshore perpetual futures all along. Traditional futures contracts provide certainty on the funding rate to contract expiration. For investors with horizons running weeks or months this is desirable. If your horizon is measured in hours or days – as must be the case with many highly levered positions – then locking in a longer-term finance rate isn't a material consideration, but nor is the inconvenience of rolling.

If investors prefer to hold an asset with leverage without having to deal with futures rolls, the normal way the market offered this has been through brokers offering margin leverage, or in some non-US markets, through a derivative known as Contracts for Difference (CfD). In both these cases, investors can hold a leveraged position from day to day, but usually with a financing cost known in advance, and with the ability to exit the position at the spot market price of the underlying asset.

Why Perpetuals At All?

Given the pre-existing, time-tested alternatives to perpetual futures, it's something of a mystery why the crypto community supported this new structure. We'd love to ask Arthur Hayes himself (now that

¹⁰ One meaningful exception is decentralized, smart-contract-based perpetual exchanges (dYdX, GMX, Drift, and similar), where funds are held on-chain rather than by a corporate counterparty that can go bankrupt.

¹¹ Duffy did not mince words: "I totally disagree with the government... I don't like to see people that don't understand products to potentially get blown out of a contract that they shouldn't be in in the first place."

President Trump's pardon has put him back in circulation) what was going through his head when BitMEX launched the perpetual swap, but absent that, we'll indulge in a bit of speculation of our own.

Here's a plausible story. The natural end users for outright, leveraged Bitcoin exposure were, overwhelmingly, people who wanted to go long. At the same time, there was a pool of sharp, arbitrage-minded capital circulating around Bitcoin markets – the same kind of capital that, in that era, was actively exploiting price gaps for Bitcoin between different exchanges and geographies (the well-documented “kimchi premium” between Korean and other exchanges is one famous example). That capital would, in principle, be happy to supply leverage to the long-only speculators – by buying spot Bitcoin and simultaneously shorting the perpetual, a textbook cash-and-carry trade – but only for a healthy return, given how uncomfortable traditional lenders (banks, prime brokers, anyone in the business of secured lending) still were about lending in *unregulated offshore markets* against Bitcoin as collateral in that period. The anticipated cost of that capital was high.

If Bitcoin were only traded on three-month rolling futures, that high financing cost would show up as a jarring, lump-sum price gap between contracts every time a trader had to roll. Say Bitcoin's current contract is trading at \$60,000, and the funding rate implied by the arbitrage capital's required return runs around 20% annualized – roughly 5% over three months, or about \$3,000. The next quarterly contract would then need to trade around \$63,000 to reflect that cost of carry. Every quarter, the long speculator rolling their position would have to visibly give up \$3,000 per contract in a single transaction – real sticker shock, and reason enough to walk away from the trade entirely.

But drip that same 20% annualized cost out continuously, eight hours at a time – working out to a bit under two basis points every eight hours – and the long speculator barely notices it happening at all. Meanwhile, the arbitrage capital on the other side earns a healthy return for supplying the leverage, paid out in a steady trickle rather than one uncomfortable lump sum.

What's more, with traditional rolling futures contracts the financing cost is primarily market-driven, and typically converges into the ballpark of risk-free rates as the market matures. But with perpetual futures, the minimum funding rate is set by formula, and ensures that short positions earn a large spread even in the presence of ample capital to finance the longs. The Economist's [Buttonwood columnist](#) recently sat down with Sam Hayes to ask about that 0.01% per-eight-hour funding rate, and came away with this take::

To a cynic, though, the funding rate conceals a bigger reason for manufacturing perps. It also includes a straightforward interest payment of 0.01% of the perp's notional value, made every eight hours, from the long side to the short. Over a year the magic of compounding ensures that this measly-sounding fee costs longs more than 10% of their position's value.¹² Needless to say, it is mostly retail traders using perps to go long and professional shorts collecting the fees. Mr Hayes is disarmingly open about this. “Basically I set it there,” he says. “I said I want to make 10% on my money...and no one's changed it since.”

Binance (and other platforms too) set 0.01% as the “base” or central rate when the futures price is within +/-0.05% of the spot price. The 0.01% every eight hours compounds to 11.6% pa – considerably higher than the current 3.5% T-bill rate, and much higher than futures on the CME are pricing in as well. To the extent a market participant thought the the perpetual futures would stay close to the spot price, they could make an attractive return by buying the CME BTC futures and selling the BTC perpetual on Binance (of course, they'd also have to be confident that the Binance platform operated as promised). Perhaps this one fact gives us much of what we need to know about the motivation and design choices of perpetual futures.

¹²An unrounded 11.6%.

While today, perpetual futures have some guardrails in place around the averaging process, the protections in offshore crypto markets against self-dealing in paint-the-tape operations are far less robust than in G7 regulated markets. Over the years, there have been frequent reports of offshore-based crypto trading firms using multiple entities to trade with each other to create desired price dynamics, without fear of regulatory oversight or penalties. Perhaps it was envisioned from the start that it would be possible to generate high funding rates by keeping the average price of the perpetual above the spot price through such machinations. Indeed, several early market participants have reported that the average funding rate over 2017, shortly after BitMEX introduced perpetual futures in mid-2016, ran at over 100%, much higher than even the 36% funding rate of 2021 reported in the table above.

It's a plausible enough explanation for why Bitcoin ended up with perpetuals instead of just leaning on the traditional futures structure that already worked fine for other assets. Of course, this is just a hypothesis – we don't actually know why perpetuals won out. What we do know is that certain arbitrage-oriented trading desks have profited handsomely (so far) from taking the other side of retail's speculative long positions, and that the "insurance funds" built from liquidated traders' excess margin have grown substantially, to the (eventual) benefit of the exchanges that run them.

Why Leverage at All?

Before wrapping up, it's worth discussing a more basic question: when does it make sense to use leverage, whether through traditional futures, perpetual futures, or a margin loan? As we discussed in our note [Leverage Sense and Non-Sense](#), the answer comes down to whether leverage raises your portfolio's expected risk-adjusted return. For people with normal levels of risk-aversion, and for what markets typically offer in terms of expected return, risk and cost of leverage, it doesn't. Leverage almost never makes sense, either, if you're sitting on free cash elsewhere in your portfolio: at that point you're simply giving away the spread between what your cash earns and what the leverage costs, for no offsetting benefit. Bitcoin funding costs, as we've shown, routinely run 5% or more over T-bills — and we find it hard to construct a scenario, under any reasonable (or even mildly unreasonable) view of Bitcoin's expected return, where it makes sense to own so much of a single volatile asset that you need to borrow to hold it.

Connecting the dots

We recognize there's a great deal more nuance here than we've covered, but we hope this at least gives a working sense of how these contracts function and relate to traditional futures contracts and other forms of leverage. Whatever you think of their merits, perpetual futures are very much part of today's financial derivatives landscape, and are already delivering exposure to stock indexes, individual equities, crude oil, gold, silver and anything else that flutters the heart and punters feel an urge to punt.¹³

¹³ For example, Hyperliquid's HIP-3 framework now hosts 24/7 perp markets on all these assets as well as an [licensed S&P 500 perpetual](#), with HIP-3 open interest hitting \$2.3B in April 2026.

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